Lecture 5

Observables and the Poisson algebra

Objectives. *Lecture 4 gave Hamilton's equations. This lecture asks how an observable — anything one can measure on a state, a function on phase space — changes along a motion. The answer is its bracket with the Hamiltonian, and the* Poisson bracket *makes the observables an algebra. The lecture proves the bracket's properties, above all the Jacobi identity; turns conservation laws into bracket equations, so that conserved quantities form a Lie algebra; computes the angular-momentum algebra $\mathfrak{so}(3)$; and describes time evolution as a motion of the observables themselves. The observables are the first piece of the course's second answer.*

5.1 Observables

Throughout Part II the phase space is an open set $\mathcal{M} \subseteq \mathbb{R}^{2n}$ with coordinates $x = (q, p) = (q_1, \dots, q_n, p_1, \dots, p_n)$, and the motion is given by Hamilton's equations (4.8), $\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$, for a Hamiltonian $H(q, p, t)$ of class C^∞ on $\mathcal{M} \times I$, I an open interval (Theorem 4.5). The pendulum lives on $\mathbb{R}^2$, with the angle read modulo 2π (§ 1.5).

Definition 5.1 (Observable)**.** An *observable* is a function $f \in C^\infty(\mathcal{M})$, real valued. A *time-dependent observable* is a C^∞ function $f(q, p, t)$ on $\mathcal{M} \times I$.

A state determines the value of every observable; an observable is a measuring device, and its reading on the state x is $f(x)$ (Figure 5.1). Observables form a real vector space and can be multiplied pointwise, and a smooth function of observables is an observable: if $f_1, \dots, f_k \in C^\infty(\mathcal{M})$ and $F \in C^\infty(\mathbb{R}^k)$, then $F(f_1, \dots, f_k) \in C^\infty(\mathcal{M})$ is the device "read $f_1, \dots, f_k$, then compute F". So $C^\infty(\mathcal{M})$ is a commutative algebra with unit 1. The examples are the coordinates q_i and p_i, the kinetic energy $|p|^2/2m$, and, for a particle in $\mathbb{R}^3$, the angular momentum $\mathbf{L} = q \times p$, whose components are three observables.

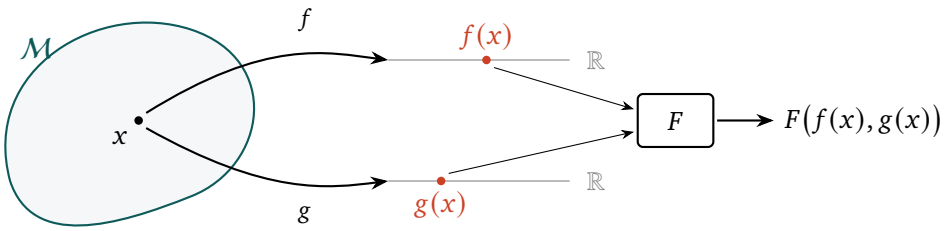


Figure 5.1: Observables as measuring devices, after Nestruev, *Smooth Manifolds and Observables*, §1. A state x gives a reading of each device; a smooth function of readings is again a device.

Some natural questions are not smooth. "Is the particle in the box B?" is the indicator function $\mathbf{1}_B(q)$, which jumps on the boundary of B. It is not an observable in the sense of

Definition 5.1: the bracket below differentiates, and $\mathbf{1}_B$ cannot be differentiated there. Its role is statistical, the probability of finding the particle in B, and Lecture 7 gives it a home by treating states as probability distributions. Smoothness is also why we take C^∞ rather than C^2: a bracket costs one derivative, and C^∞ is the class that brackets never leave.

5.2 The Poisson bracket

How does the reading of an observable change along a motion? Let $x(t) = (q(t), p(t))$ solve Hamilton's equations. By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(x(t),t\big) = \sum_{i=1}^{n}\Big(\frac{\partial f}{\partial q_i}\dot q_i + \frac{\partial f}{\partial p_i}\dot p_i\Big) + \frac{\partial f}{\partial t} = \sum_{i=1}^{n}\Big(\frac{\partial f}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q_i}\Big) + \frac{\partial f}{\partial t}.$$

The sum is built from f and H alone, and it deserves a name.

Definition 5.2 (Poisson bracket)**.** For $f, g \in C^\infty(\mathcal{M})$,

$$\{f,g\} = \sum_{i=1}^{n}\Big(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i}\Big). \tag{5.1}$$

For time-dependent observables the bracket is taken at each fixed t.

So, along every solution of Hamilton's equations,

$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(x(t),t\big) = \Big(\{f,H\} + \frac{\partial f}{\partial t}\Big)\big(x(t),t\big). \tag{5.2}$$

This is the sign convention of the whole collection.[1] In the notation of (4.9), $\{f,H\} = \nabla f \cdot J\nabla H$ is the derivative of f along the velocity field $J\nabla H$ of the motion: *it measures how fast the motion cuts across the level sets of* f, and it vanishes where the motion is tangent to them (Figure 5.2).

The coordinates themselves give the simplest brackets. Since $\partial q_i/\partial q_j = \delta_{ij}$ and $\partial q_i/\partial p_j = 0$,

$$\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij}, \tag{5.3}$$

the *canonical relations*, and for every observable f

$$\{f, p_i\} = \frac{\partial f}{\partial q_i}, \qquad \{f, q_i\} = -\frac{\partial f}{\partial p_i}\,: \tag{5.4}$$

brackets with the coordinates are partial derivatives. With $f = q_i$ and $f = p_i$ in (5.2), Hamilton's equations become

$$\dot q_i = \{q_i, H\}\,, \qquad \dot p_i = \{p_i, H\}\,. \tag{5.5}$$

For the oscillator $H = p^2/2m + \frac{1}{2}m\nu^2 q^2$ of Lecture 4, $\{q,H\} = p/m$ and $\{p,H\} = -m\nu^2 q$.

For the pendulum $H = \frac{1}{2}p^2 + 1 - \cos q$ and $f = \frac{1}{2}(q^2+p^2)$,

$$\{f,H\} = \frac{\partial f}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial H}{\partial q} = qp - p\sin q = p\,(q - \sin q).$$

Since $q - \sin q$ has the sign of q, $\{f,H\} > 0$ exactly where $qp > 0$: in the first and third quadrants the motion crosses the circles outward, in the second and fourth inward (Figure 5.2). The distance from the origin is not conserved, as the non-circular level sets of H show.

[1] Faddeev and Yakubovskii, and Takhtajan, write $\sum(\partial_p f\,\partial_q g - \partial_q f\,\partial_p g)$, the negative of (5.1); their $\dot f = \{H,f\}$ is our $\dot f = \{f,H\}$.

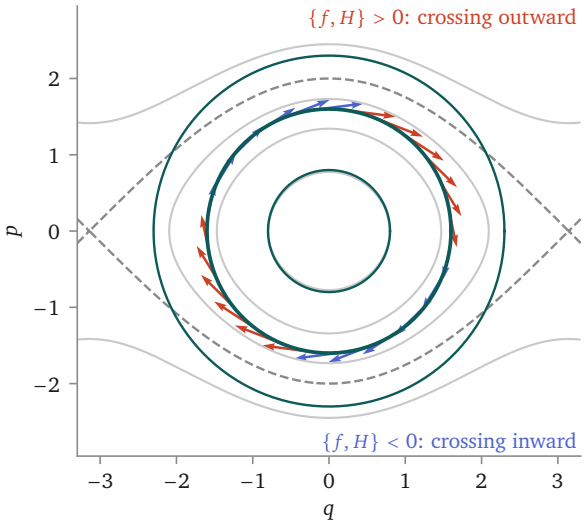


Figure 5.2: The pendulum $H = \frac{1}{2}p^2 + 1 - \cos q$ (grey level sets; dashed: the separatrix) and the circles $f = \frac{1}{2}(q^2+p^2) = \text{const}$ (teal). On one circle the arrows are the velocity field $J\nabla H = (p, -\sin q)$. Since $\{f, H\} = p(q - \sin q)$, they point outward where it is positive, inward where it is negative, and along the circle where it vanishes.

5.3 The algebra

Proposition 5.3 (Bilinear, antisymmetric, Leibniz)**.** *For observables f, g, h, real a, b and $F \in C^\infty(\mathbb{R}^k)$:*

1. $\{f, ag + bh\} = a\{f, g\} + b\{f, h\}$;
2. $\{g, f\} = -\{f, g\}$; *in particular* $\{f, f\} = 0$;
3. *(Leibniz)* $\{f, gh\} = \{f, g\}\, h + g\, \{f, h\}$, *and more generally* $\{f, F(g_1, \dots, g_k)\} = \sum_j \partial_j F(g_1, \dots, g_k)\, \{f, g_j\}$.

Proof. (5.1) is linear in g and changes sign when f and g are exchanged. For (3), the chain rule gives $\partial F(g_1, \dots, g_k)/\partial p_i = \sum_j \partial_j F\, \partial g_j/\partial p_i$, and the same in q_i; insert in (5.1). The product rule is the case $F(u, v) = uv$. ∎

By (2), (1) and (3) hold in the first slot too. Fix f and write $D_f = \{f, \cdot\}$, that is,

$$D_f = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q_i}\frac{\partial}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial}{\partial q_i} \right),$$

a *first-order differential operator*: an operator $D = \sum_k a_k(x)\, \partial/\partial x_k$ in the $2n$ variables x, with C^∞ coefficients. Leibniz says that D_f is a derivation of the product. The bracket is not associative: the third property, the one that carries the mechanics, relates the three ways of bracketing f, g, h.

Lemma 5.4 (Commutators of first-order operators)**.** *If $D = \sum_k a_k \partial_k$ and $E = \sum_k b_k \partial_k$ are first-order operators, then $[D, E] = DE - ED$ is the first-order operator*

$$[D, E] = \sum_k \left(Db_k - Ea_k \right) \partial_k .$$

Proof. For $u \in C^\infty$, $Du = \sum_{j,k} a_j \partial_j (b_k \partial_k u) = \sum_k (Db_k)\partial_k u + \sum_{j,k} a_j b_k \, \partial_j \partial_k u$. Subtract the same with a and b exchanged. The second-order terms $\sum_{j,k}(a_j b_k - b_j a_k)\partial_j \partial_k u$ cancel, because $\partial_j \partial_k u = \partial_k \partial_j u$. ∎

Theorem 5.5 (The Jacobi identity). *For all observables f, g, h,*

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0. \tag{5.6}$$

Proof. Expand the left side by (5.1). The coefficients in (5.1) are constants, so each term of the expansion is a product of first derivatives of two of the three functions and one second derivative of the third. Collect the terms that contain a second derivative of h. The third bracket $\{h, \{f, g\}\}$ contains only first derivatives of h, so these terms all come from

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} = \{f, \{g, h\}\} - \{g, \{f, h\}\} = (D_f D_g - D_g D_f)h,$$

and they are its second-order part. By Lemma 5.4 that part vanishes. The cyclic sum is unchanged when f, g, h are permuted cyclically, so the same argument removes the terms with a second derivative of f and those with a second derivative of g. No term is left. ∎

The argument replaces a long expansion by a count. By antisymmetry (5.6) can be rewritten in two useful forms:

$$\{f, \{g, h\}\} = \{\{f, g\}, h\} + \{g, \{f, h\}\}, \qquad \{\{f, g\}, H\} = \{\{f, H\}, g\} + \{f, \{g, H\}\}. \tag{5.7}$$

The first says that D_f is a derivation of the bracket, as Leibniz says it is a derivation of the product. The second says that the rate $\{\cdot, H\}$ obeys a product rule for brackets; it is the form used below.

Remark 5.6 (A non-constant bracket). Only one step used the constant coefficients: that every term has a second derivative. For a bracket $\{f, g\} = \sum_{a,b} \Pi_{ab}(x)\, \partial_a f \, \partial_b g$ with a C^∞ antisymmetric matrix $\Pi(x)$, the second-order terms still cancel (each $\{f, \cdot\}$ is still first order), but terms with derivatives of Π remain. They are trilinear in first derivatives, so the cyclic sum equals $\sum_{a,b,c} \partial_a f \, \partial_b g \, \partial_c h$ times the cyclic sum for the coordinate functions x_a, x_b, x_c (whose second derivatives vanish): Jacobi holds for all observables if and only if it holds for the coordinates. Lecture 6 uses this.

Definition 5.7 (Poisson algebra). A *Poisson algebra* is a real vector space $\mathcal{A}$ with a commutative, associative product with unit and a bilinear bracket $\{\,,\,\}$ that is antisymmetric, satisfies the Jacobi identity, and satisfies Leibniz: $\{f, gh\} = \{f, g\}\, h + g\, \{f, h\}$.

A bilinear, antisymmetric bracket satisfying Jacobi is a *Lie algebra*. By Proposition 5.3 and Theorem 5.5, $C^\infty(\mathcal{M})$ with (5.1) is a Poisson algebra; as a Lie algebra it is infinite-dimensional, the *Poisson–Lie algebra of observables*. A Poisson algebra has two products tied by Leibniz: the product is how devices combine, and the bracket is how they change in time.

Proposition 5.8 (The centre). *Let $\mathcal{M}$ be connected. If $\{f, g\} = 0$ for every $g \in C^\infty(\mathcal{M})$, then f is constant.*

Proof. Take $g = p_i$ and $g = q_i$ in (5.4): $\partial f/\partial q_i = \partial f/\partial p_i = 0$ for every i. A function with zero gradient on a connected open set is constant. ∎

Conversely, constants commute with everything, so on a connected $\mathcal{M}$ the *centre* $\{f : \{f, \cdot\} = 0\}$ is $\mathbb{R} \cdot 1$. On a disconnected $\mathcal{M}$, a function that is constant on each component is also central.

5.4 Conserved quantities

Definition 5.9 (Conserved quantity)**.** A time-dependent observable f is *conserved* (an *integral of the motion*) if $f(x(t),t)$ is constant along every solution of Hamilton's equations in $\mathcal{M}\times I$.

Proposition 5.10 (Conservation is a bracket equation · *proved; "only if" uses T1 (cited)*)**.** *f is conserved if and only if*

$$\frac{\partial f}{\partial t}+\{f,H\}=0 \quad \text{on } \mathcal{M}\times I. \tag{5.8}$$

For f without explicit t this reads $\{f,H\}=0$. If H does not depend on t, then H is conserved.

Proof. By (5.2) the derivative of $f(x(t),t)$ along a solution is the left side of (5.8) at $(x(t),t)$, so (5.8) implies conservation. Conversely, through every $(x_0,t_0)\in\mathcal{M}\times I$ passes a solution (Theorem 1.3, cited, applied to (1.6)); at t_0 its derivative is the left side at (x_0,t_0), and it vanishes. For $f=H$, $\{H,H\}=0$. ∎

A conserved quantity can depend on t explicitly, and still be conserved.

Example 5.11 (The boost)**.** For a free particle in $\mathbb{R}^3$, $H=|p|^2/2m$, take the vector $\mathbf{K}=mq-tp$, which depends on the time explicitly. Check (5.8) componentwise: $\partial_t K_i=-p_i$, and $\{K_i,H\}=m\{q_i,H\}=p_i$ by (5.5); the sum vanishes. Its value on the motion $q(t)=q_0+tp/m$ is $\mathbf{K}=mq_0$: m times the initial position. Its brackets with the momenta are

$$\{K_i,p_j\}=m\,\delta_{ij}, \tag{5.9}$$

by (5.3).

Theorem 5.12 (Poisson · *proved; uses Proposition 5.10, hence T1 (cited)*)**.** *If f and g are conserved, so is $\{f,g\}$. The conserved quantities form a Lie subalgebra of the time-dependent observables, and they are also closed under products and smooth functions.*

Proof. The bracket is taken at fixed t, and f,g are C^∞, so $\partial_t\{f,g\}=\{\partial_t f,g\}+\{f,\partial_t g\}$. With the second form of (5.7),

$$\frac{\partial}{\partial t}\{f,g\}+\{\{f,g\},H\}=\left\{\frac{\partial f}{\partial t}+\{f,H\},\ g\right\}+\left\{f,\ \frac{\partial g}{\partial t}+\{g,H\}\right\}=0$$

by Proposition 5.10. Products and smooth functions of quantities that are constant along each motion are constant along each motion. ∎

Poisson's theorem may produce a new integral, or nothing new. For the boost, $\{K_i,p_j\}=m\delta_{ij}$ is a constant: conserved, but useless. For a central force all three components of $\mathbf{L}=q\times p$ are conserved (Lecture 3), and $\{L_x,L_y\}=L_z$ (§ 5.5) returns a quantity already known.

5.5 Angular momentum

Let $q,p\in\mathbb{R}^3$, write ε_{abc} for the alternating symbol, and $L_a=\sum_{j,k}\varepsilon_{ajk}q_jp_k$ for the components of $\mathbf{L}=q\times p$. A triple u of observables is a *vector* if $\{L_a,u_b\}=\sum_c\varepsilon_{abc}u_c$, and an observable s is a *scalar* if $\{L_a,s\}=0$, for all a,b. For constant $n\in\mathbb{R}^3$, componentwise,

$$\{n\cdot\mathbf{L},u\}=u\times n,\qquad \{n\cdot\mathbf{L},s\}=0. \tag{5.10}$$

Proposition 5.13 (L rotates vectors). *q and p are vectors. If u, w are vectors and $s, s_1, \dots, s_k$ are scalars, then $u \cdot w$ and $F(s_1, \dots, s_k)$ (F smooth on an open set containing the values of $(s_1, \dots, s_k)$) are scalars, and $u \times w$, $u + w$ and su are vectors.*

Proof. By Leibniz and (5.3), $\{L_a, q_b\} = \sum_{j,k} \varepsilon_{ajk} q_j \{p_k, q_b\} = -\sum_j \varepsilon_{ajb} q_j = \sum_c \varepsilon_{abc} q_c$, and $\{L_a, p_b\} = \sum_{j,k} \varepsilon_{ajk} \{q_j, p_b\} p_k = \sum_c \varepsilon_{abc} p_c$. By Leibniz and (5.10), $\{n \cdot \mathbf{L}, u \cdot w\} = (u \times n) \cdot w + u \cdot (w \times n) = 0$, since both triple products equal $\pm n \cdot (w \times u)$ with opposite signs, and

$$\{n \cdot \mathbf{L}, u \times w\} = (u \times n) \times w + u \times (w \times n) = (u \times w) \times n$$

by the Jacobi identity of the cross product. The rest is linearity and Proposition 5.3(3). ∎

Hence $\mathbf{L} = q \times p$ is a vector:

$$\{L_a, L_b\} = \sum_c \varepsilon_{abc} L_c, \qquad \text{e.g. } \{L_x, L_y\} = L_z. \tag{5.11}$$

By Poisson's theorem, *if two components of* $\mathbf{L}$ *are conserved, so is the third*. The scalars include $|q|^2$, $|p|^2$, $q \cdot p$, $|\mathbf{L}|^2$, and, on $q \neq 0$, every smooth function of $|q|$. So every $H = |p|^2/2m + V(|q|)$ with $V \in C^\infty(0, \infty)$ is a scalar, and all three L_a are conserved, by Proposition 5.10.

5.6 The Heisenberg picture

So far the state moved and the observable was fixed. Now let H not depend on t, with a complete flow φ_t (Definition 1.8), as for $\frac{1}{2}|p|^2 + V(q)$ with $V \in C^\infty(\mathbb{R}^n)$ bounded below (Proposition 1.9). The *Heisenberg picture* keeps the state fixed and moves the observable: $f_t := f \circ \varphi_t$, so $f_t(x)$ is the reading of f at time t on the motion that starts at x.

Theorem 5.14 (Theorem A: the Heisenberg equation · *proved modulo T2 (cited)*). *Let $f \in C^\infty(\mathcal{M})$.*

1. *$(t, x) \mapsto f_t(x)$ is C^∞, and*

$$\frac{\partial f_t}{\partial t} = \{f_t, H\} = \{f, H\} \circ \varphi_t. \tag{5.12}$$

2. *(Uniqueness) If $F \colon \mathbb{R} \times \mathcal{M} \to \mathbb{R}$, $(t, x) \mapsto F_t(x)$, is C^1, with $\partial_t F_t = \{F_t, H\}$ and $F_0 = f$, then $F_t = f_t$ for all t.*

Proof. (1) The flow is C^∞ in (t, x) (Theorem 1.5), and by the group law (1.7) $f_{t+s} = f_t \circ \varphi_s = f_s \circ \varphi_t$. Differentiate at $s = 0$: the curve $s \mapsto \varphi_s(y)$ has velocity $J\nabla H(y)$ at $s = 0$, so the chain rule turns the two expressions into $\nabla f_t \cdot J\nabla H(x) = \{f_t, H\}(x)$ and $\{f, H\}(\varphi_t x)$. Both equal $\partial_t f_t(x)$.

(2) Fix t and x, and let $G(s) = F_{t-s}(\varphi_s(x))$ for s between 0 and t. By the chain rule,

$$G'(s) = -\big(\partial_t F\big)_{t-s}(\varphi_s x) + \{F_{t-s}, H\}(\varphi_s x) = 0.$$

So $F_t(x) = G(0) = G(t) = F_0(\varphi_t x) = f_t(x)$. ∎

The theorem separates two meanings of "$\mathrm{d}f/\mathrm{d}t$". Along the motion $x(t) = \varphi_t x$ the *number* $f(x(t))$ changes at the rate $\{f, H\}(x(t))$ (5.2); in the Heisenberg picture the *function* f_t changes at the fixed point x at the rate $\{f_t, H\}(x)$. By (5.12) the two rates are equal: evolve-then-bracket equals bracket-then-evolve, for the bracket with H. For any two observables this is Theorem B (Theorem 6.4), and it rests on Jacobi. The level sets of f_t are those of f carried *backwards*: $\{f_t = c\} = \varphi_{-t}(\{f = c\})$ (Figure 5.3).

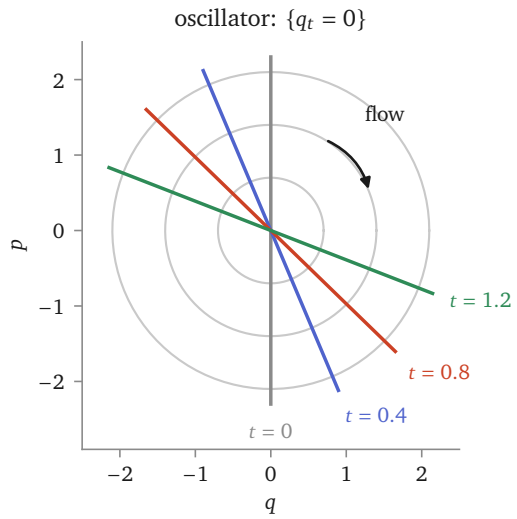


Figure 5.3: $\{q_t = 0\} = \varphi_{-t}(\{q = 0\})$, the states that reach $q = 0$ after time t, for the oscillator, $m = \nu = 1$: states turn clockwise, the line $q\cos t + p\sin t = 0$ counter-clockwise.

Example 5.15 (The oscillator). Let $H = p^2/2m + \frac{1}{2}m\nu^2 q^2$; its flow is complete, since Hamilton's equations are linear. Its observables q_t and p_t are

$$q_t = q\cos\nu t + \frac{p}{m\nu}\sin\nu t, \qquad p_t = p\cos\nu t - m\nu q\sin\nu t,$$

for every t: each right side is C^∞, equals q (resp. p) at $t = 0$ and solves $\partial_t F_t = \{F_t, H\}$, as one checks directly, so Theorem 5.14(2) identifies it with q_t (resp. p_t). The state $(1, 0)$ moves to $(\cos\nu t, -m\nu\sin\nu t)$: clockwise, the sign fact that § 6.1 records.

5.7 Finite-dimensional subalgebras

$C^\infty(\mathcal{M})$ is infinite-dimensional, but polynomials in (q, p) contain small Lie algebras. The reason is a count of degrees.

Proposition 5.16 (Degree counting). *If f and g are polynomials in (q, p) of degrees a and b, then $\{f, g\}$ is a polynomial of degree at most $a + b - 2$. If f and g are homogeneous, $\{f, g\}$ is homogeneous of degree $a + b - 2$, or zero.*

Proof. Each term of (5.1) is a product of a derivative of f, of degree at most $a - 1$, and a derivative of g, of degree at most $b - 1$. ∎

So the polynomials of degree at most 2 form a Lie subalgebra, and inside it:

- *The Heisenberg algebra* $\mathfrak{h}_n = \operatorname{span}\{1, q_1, \dots, q_n, p_1, \dots, p_n\}$, of dimension $2n + 1$, with
$$\{q_i, p_j\} = \delta_{ij}\,1, \qquad \{q_i, q_j\} = \{p_i, p_j\} = 0, \qquad \{1, \cdot\} = 0. \tag{5.13}$$
Its centre is $\mathbb{R}\cdot 1$, and the constant 1 appears as a bracket.
- *The homogeneous quadratics*, of dimension $n(2n + 1)$: closed, since $2 + 2 - 2 = 2$. For $n = 1$, a basis is qp, $p^2/2$, $q^2/2$, with
$$\{qp, p^2/2\} = p^2, \qquad \{qp, q^2/2\} = -q^2, \qquad \{q^2/2, p^2/2\} = qp.$$
A quadratic bracketed with a linear function is linear ($2 + 1 - 2 = 1$), so the quadratics act on $\operatorname{span}\{q_i, p_i\}$ by linear maps. Lecture 6 identifies this Lie algebra with $\mathfrak{sp}(2n)$.

- $\mathfrak{so}(3)$*:* for $n = 3$ the components L_a are homogeneous quadratics, and (5.11) says that their span is a subalgebra with the bracket of $(\mathbb{R}^3, \times)$.

Degree 3 is not closed: $\{q^3, p^3\} = 9q^2p^2$, and repeated brackets climb in degree. The polynomials of degree at most 2 are where classical and quantum brackets can be matched exactly (Lecture 12).

Summary.

- *Handed on:* observables and the bracket, with $\mathrm{d}f/\mathrm{d}t = \{f, H\} + \partial_t f$ along motions (Definition 5.2); Jacobi, so that $C^\infty(\mathcal{M})$ is a Poisson algebra (Theorem 5.5); conservation as a bracket equation, and conserved quantities form a Lie algebra (Proposition 5.10 and Theorem 5.12); $\{L_a, L_b\} = \varepsilon_{abc}L_c$; Theorem A, $f_t = f \circ \varphi_t$ (Theorem 5.14); the Heisenberg algebra and the quadratics (Proposition 5.16).
- *Next:* Lecture 6 asks what flow each observable generates; Jacobi will say that these flows preserve the bracket.

References

Faddeev and Yakubovskii, *Lectures on Quantum Mechanics for Mathematics Students*, §1 (pp. 1–6), and Arnold, *Mathematical Methods of Classical Mechanics*, §40 (pp. 214–217): the proof of Jacobi by counting second derivatives is theirs. Takhtajan, *Quantum Mechanics for Mathematicians*, Ch. 1, §2.4 (pp. 38–39) and §2.6 (pp. 46–47). Tong, *Lectures on Classical Dynamics*, §4.3 (pp. 94–96).